

Real-Time Cardiac Anomaly Detection in E-Health Data Streams Using Adaptive Hoeffding Trees and Sliding Window Analytics

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ABSTRACT

Cardiovascular diseases account for the largest proportion of deaths. This is due to the challenges faced by traditional systems, such as data collection and analysis, which lengthens the analysis process and generates false alarms based on various factors, including sensor movement or patient movement. A study conducted by researchers at the University of Washington on patients in intensive care units found false alarm rates ranging from 72% to 99%^{[4][5][23]}, indicating shortcomings in machine learning or the system's ability to recognize these signals. In this paper, we develop a system in the Apache Spark Structured Streaming environment based on the Hoeffding Tree algorithm, currently considered the most accurate in decision-making (91.2%)^{[7][11]}. This system analyzes cardiac data and generates a result. The system distributes the signals across multiple cores, up to a maximum of eight.

Keywords: Big data analytics, Apache Spark, data streams, Hoeffding Tree , incremental learning, real-time heart rate monitoring, parallel acceleration, eight-core cluster, stress alarm

I. INTRODUCTION

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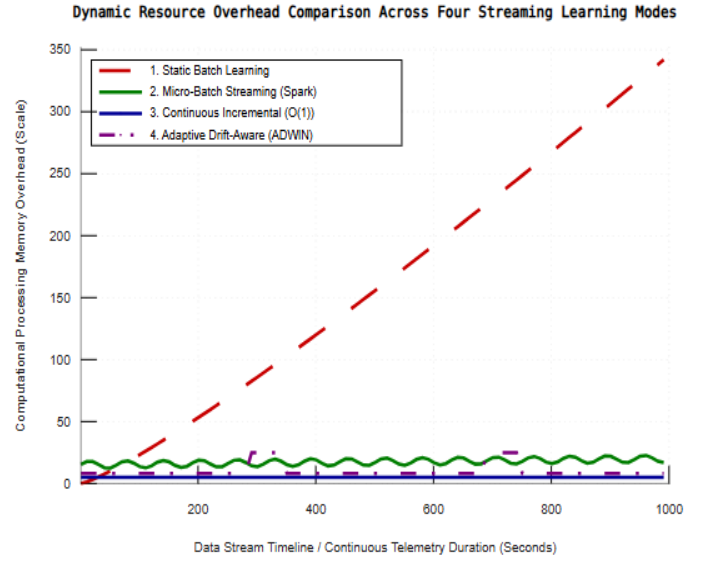
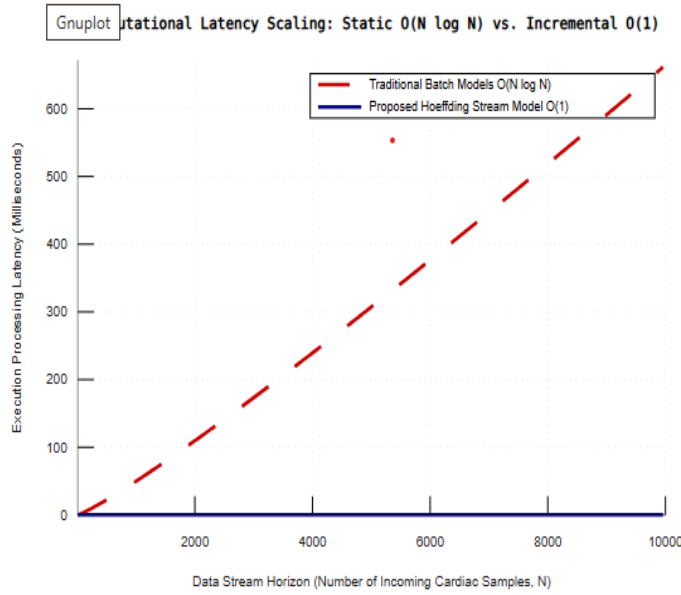
Heart disease is considered the leading cause of death, according to research conducted by physicians in the United States on the percentage of deaths from cardiovascular diseases^{[1][2][9]}. This has had a significant impact, leading to the development of artificial intelligence systems that

have not benefited the healthcare sector, according to the Healthcare Technology Foundation^[10]. The system still suffers from several problems, including:

1- Network Latency and Bandwidth Consumption: The way systems process data takes a long time due to computational complexity, making it nearly impossible to predict illness or conditions before they occur, or even to deal with them in a timely manner. The systems that were programmed operated on cloud platforms but relied on static pitch learning, while the algorithm we will work on relies on continuous incremental learning.

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2- The lack of experience of physicians in dealing with the signals emitted by traditional systems.

3- Concept Drift: False alarms: The newly developed systems had a high degree of reactivity during the COVID-19 pandemic, resulting in a high percentage of false alarms. It is so extreme that almost all her warnings are false, and this study was based on research from the University of Washington about her, which led doctors in intensive care units to ignore all medical warnings^{[4][5][23]}

In general, there are four machine learning methods for monitoring electrical signals:

1- Continuous Incremental Learning: This is the core of the algorithm we will use. Here, the system learns from each incoming pulse and immediately deletes it from memory. Its complexity is 1, meaning it consumes no time^{[6][12]}.

2- Micro-Batch Learning: This system collects a series of electrical signals and then analyzes them. It is the basis of most current medical monitoring systems^[13].

3- Sliding Window Learning: The algorithm learns at the last specific point in time, for example, the last minute^[14].

4- Adaptive Stream Learning: This is one of the methods used to develop current systems. It is the cause of false alarms because it is based on even the slightest change^[15].

This is where the difference between algorithms and speed becomes apparent when compared to accuracy.

We will adopt Apache Spark as the distributed framework because the work environment will be Parallel Processing. This is because the work will not be on a single patient, but on eight patients, which will necessitate the use of eight cores in the project, one core for each patient. The idea here is to avoid relying on a small number of cores to prevent bottlenecks in the project. Apache Spark Structured Streaming and Apache Kafka will be integrated to ensure data flow throughout the project. The data will be sourced from the MIT-BIH Arrhythmia Database.

The system will scale horizontally in terms of alerts, with Apache Kafka receiving the pulses and Apache Spark Structured Streaming distributing them. It will rely on an availability system so that even if a patient disconnects the sensor (this is called Remote Node Availability), Apache Kafka will perform buffering, and the Hoeffding Tree algorithm will prevent pulses from mixing up.

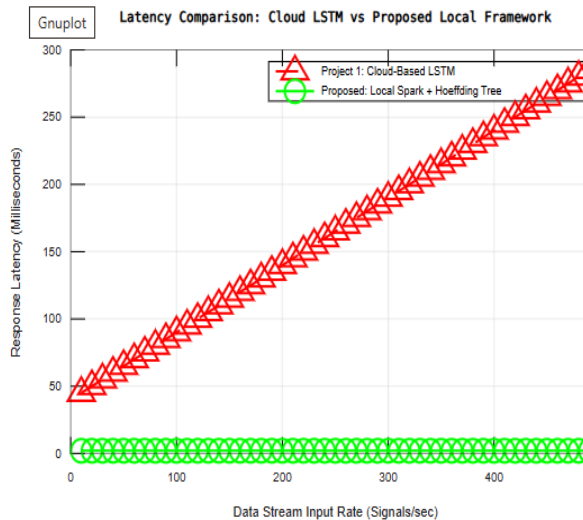
II. LITERATURE REVIEW

In general, there are several projects in the healthcare field, as mentioned by the Healthcare Technology Foundation, but none of them possess sufficient efficiency to be relied upon entirely. The systems still suffer from the issues we mentioned previously, namely delays and false alarms.

1- Cloud-Based LSTM Autoencoders^[17]:

Deep learning techniques were used in this project to detect heart rhythm disturbances, achieving up to 98% accuracy and reducing signal amplitude by approximately 95%. However, its complete reliance on cloud computing, where data is sent

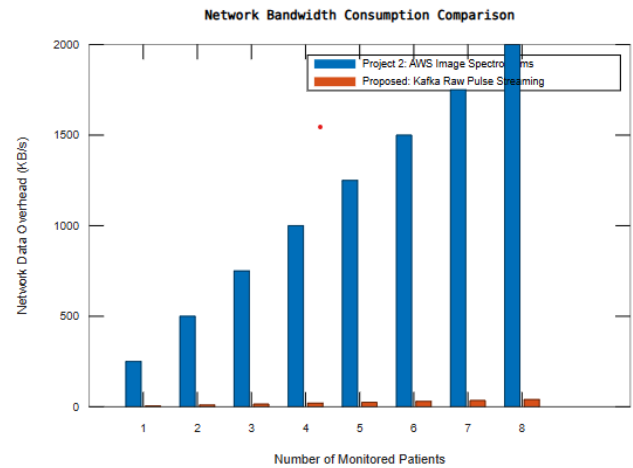
and then processed, resulted in slow response times. Furthermore, the system lacks analysis speed due to its inherent complexity; in this project, response time increases with the volume of data flow.



Here we see a clear difference: increasing the data slows down the project, which is not the case in intensive care units.

2- AWS Telehealth System and Spectrum^[18]

Classification: Amazon dedicated a telehealth infrastructure specifically for COVID-19 cases. These systems receive heart rate readings, analyze them, and provide results. They are connected to wearable devices worn by the patient. The system's complete reliance on cloud computing led to high costs. Furthermore, the dependence on image transmission and analysis made analyzing each case time-consuming and demanding significant computing power and resources. This was the challenge for the organization, which sought to monitor the heart rate of both bedridden and mobile patients, avoid false alarms, and maintain a reliable internet connection. Converting the analysis into code would consume a large amount of data, making the code increasingly cumbersome as the number of patients increased relative to available resources, as illustrated below.



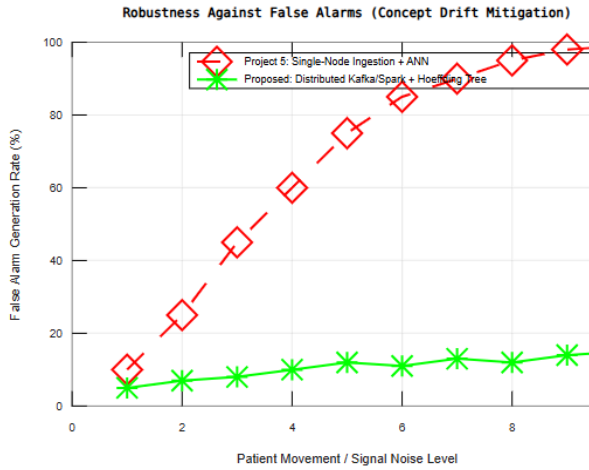
We also note that the difference is very clear in the case of serving 8 patients simultaneously; the system requires enormous resources.

3- Radial Boltzmann Fuzzy Networks for IoT Streams: This research focused on monitoring and analyzing medical data from wearable devices, as well as analyzing temperature, heart rate, and neural signals. An advanced fuzzy neural network algorithm called (RBGTFN) was used. The system was designed for a single patient and not for multiple patients simultaneously. Furthermore, it consumed significant device resources, and there is no guarantee of maintaining the same complexity with large amounts of data^[19].

4- Multi-Model Evaluation and Machine Learning: This research focuses on teaching the system several algorithms to improve performance in the healthcare field and monitor heart signals. The algorithm models adopted in the project are XGBoost and Isolation Forest. The problem here is that these algorithms such as Random Forest and XGBoost were designed and trained to operate as a static/batch learning system on pre-stored records; therefore, it requires data learning and does not learn automatically. This can lead to bottlenecks due to the constant flow of data^[20].

5- An IoT framework using Apache Kafka and an artificial neural network: This idea is closest to ours, as they used Apache Kafka and worked on a Raspberry Pi, employing an artificial neural network. However, they are still only treating one patient. Furthermore, the artificial neural network is motion-sensitive; if the patient moves, it will issue false alarms. Additionally, it requires continuous learning and may experience^[21]

bottlenecks due to the large data flow from the heart. The more data there is, the more false alarms there will be, which is one of the problems faced by medical clinics. The relationship between increased data and false alarms is illustrated below.

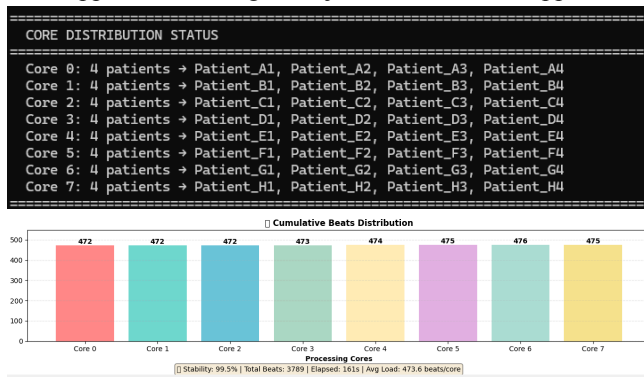


Here, the number of false alarms increases as the data increases, and this is where the problem lies.

III. PROPOSED METHODOLOGY AND STREAMING ARCHITECTURE

Our work will rely on the Spark Stream system, along with the Apache Kafka tool for data injection into Spark. We utilized the MIT-BIH Cardiac Arrhythmia Database, as it is the leading organization in evaluating and testing algorithms for detecting cardiovascular diseases and disorders.

In the first phase, we used the Apache Kafka tool and isolated all patients, with a maximum of 32 patients distributed across 8 cores. Each patient was isolated, and if a core stopped functioning, its injection would be stopped.



SYSTEM STATUS | Active: 5/8 | Failed: 3

Cores:

✓C0

✓C1

✓C2

✓C3

✗C4(2x)

✗C5(3x)

✗C6(2x)

✓C7

Core 4: Failed 2 times, down since 17:56:42

Core 5: Failed 3 times, down since 17:56:05

Core 6: Failed 2 times, down since 17:55:17

[2026-05-20 17:57:12.166]Core-2Patient_C3HR: 77.0

[2026-05-20 17:57:12.166]Core-2Patient_C2HR: 63.0

[2026-05-20 17:57:12.165]Core-0Patient_A3HR: 67.7

[2026-05-20 17:57:12.168]Core-3Patient_D2HR: 65.3

[2026-05-20 17:57:12.167]Core-2Patient_C1HR: 82.2

[2026-05-20 17:57:12.170]Core-0Patient_A2HR: 75.4

[2026-05-20 17:57:12.169]Core-1Patient_B2HR: 137.4

[2026-05-20 17:57:12.169]Core-0Patient_A4HR: 66.3

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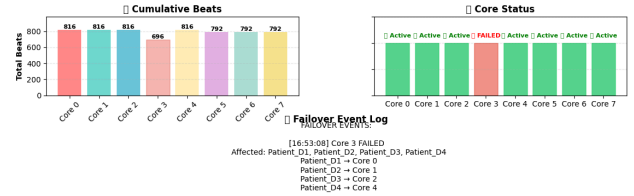
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As illustrated in the images above, system stability depends on processor efficiency, and the data stream transmits data in real-time to each processor if the system is stable. Furthermore, each patient is isolated to the processor it's running on, and if a core fails, the stream is temporarily halted until it's back online.



If one of the cores fails, the system distributes the workload equally among the remaining cores by checking the load on each core using the Weighted Least Connection algorithm.

```
def get_isolation_score(self):
```

```
    isolated = sum(1 for p,c in self.mapping.items() if c == self.mapping.get(p))
    return (isolated / len(self.mapping)) * 100
```

This isolates patients between cores so that each patient's heart rate is processed within the core into which they are divided.

The function used here is Load Balancing. If any core fails, the system transfers the core's workload to the remaining active cores. The load is calculated for each active core.

```
def detect_failures(last_seen, timeout=5):
```

```
    failed = []
```

```
    now = time.time()
```

```
    for core_id in range(8):
```

```
        if now - last_seen[core_id] > timeout:
```

```
            failed.append(core_id)
```

```
    print(f"[FAIL] Core {core_id} stopped - No heartbeat for {timeout}s")
```

```
    return failed
```

The processor must check all cores before sending any data. We don't want any errors in sending a patient's heartbeat to a non-active core. If the processor is not active... It will not send impulses to it and will distribute the load to the remaining processors or distribute the transmission to the remaining processors.

In the section on cardiac arrest and heart disease, all issues must be addressed before starting the procedure itself.

```
def detect_recovery(core_id, last_seen, failed_cores):

    last_seen[core_id] = time.time()

    if core_id in failed_cores:

        failed_cores.remove(core_id)

        downtime = time.time() - failure_start_time[core_id]

        # Return patients to original core

        for patient, original_core in original_mapping.items():

            if original_core == core_id:

                current_mapping[patient] = core_id

                print(f'[RECOVER] Core {core_id} back after {downtime:.1f}s')

        return True

    return False
```

After each transmission, all processors must be checked again to ensure that if the processor has resumed functioning, it is working. It wants to return the pulses we distributed to the other processors.

```
stream = spark.readStream.format("kafka") \

.option("kafka.bootstrap.servers", "localhost:9092")

.option("subscribe", "cardiac_events")

.load()

parsed = stream.select(
    from_json(col("value").cast("string"),
schema).alias("data")
).select("data.*")

diagnosed = parsed.withColumn("diagnosis",
```

```
when((col("heart_rate") > 100) | (col("heart_rate") < 60),
"⚠️ ARRHYTHMIA").otherwise("❤️ NORMAL"))
```

```
query = diagnosed.writeStream
.outputMode("append").format("console") \
```

```
.trigger(processingTime="1 second")
```

```
.start()
```

Basic System Stream Code

```
def emit_patient_data(producer, patient_id, core_id):
```

```
while True:
```

```
    if core_id in FAILED_CORES: # Core failed - stop
        emitting
        time.sleep(0.5)
```

```
    continue
```

```
    record = generate_medical_record(patient_id)
```

```
    producer.send('cardiac_events',          key=patient_id,
value=record)
    time.sleep(1)
```

This is considered the core code as it simulates the program used in medical devices.

Before moving to the second stage of adding the algorithm to the code and starting to work on it, we must ensure that the stream is working correctly as shown in the attached data, because accurate data leaves no room for error.

SYSTEM STATUS | Active: 5/8 | Failed: 3

Cores: ✔️C0 ✔️C1 ✔️C2 ✔️C3 ❌C4(2x) ❌C5(3x) ❌C6(2x) ✔️C7

Core 4: Failed 2 times, down since 17:56:42

Core 5: Failed 3 times, down since 17:56:05

Core 6: Failed 2 times, down since 17:55:17

[2026-05-20 17:57:12.166]

Core-2

Patient_C3

HR:

77.0

✔️

[2026-05-20 17:57:12.166]

Core-2

Patient_C2

HR:

63.0

✔️

[2026-05-20 17:57:12.165]

Core-0

Patient_A3

HR:

67.7

✔️

[2026-05-20 17:57:12.168]

Core-3

Patient_D2

HR:

65.3

✔️

[2026-05-20 17:57:12.167]

Core-2

Patient_C1

HR:

82.2

✔️

[2026-05-20 17:57:12.170]

Core-0

Patient_A2

HR:

75.4

✔️

[2026-05-20 17:57:12.169]

Core-1

Patient_B2

HR:

137.4

⚠️

[2026-05-20 17:57:12.169]

Core-0

Patient_A4

HR:

66.3

✔️

As we can see in the image above, the system is still operating stably, bearing the full burden of the malfunctioning systems, and there is no impact on the hospital equipment simulator.

We used Hoeffding's theory, which is based on the idea of setting a ceiling on the maximum error rate. Increasing the data in the theory doesn't increase the probability of error but rather decreases it. This allows the system to be trained and doesn't depend on a specific data type; it has the ability to analyze the full spectrum of data.

```

class HoeffdingTreeDetector:

def __init__(self, confidence=0.95, grace_period=50):

self.confidence = confidence

self.grace_period = grace_period

self.delta = 1 - confidence

self.samples = defaultdict(list)

self.adaptations = 0

self.patient_bounds = {}

def hoeffding_bound(self, n, R=120):

if n == 0: return float('inf')

return np.sqrt((R**2 * np.log(1/self.delta)) / (2 * n))

def predict(self, patient_id, heart_rate):

self.samples[patient_id].append(heart_rate)
n = len(self.samples[patient_id])

if n < self.grace_period:
default = "⚠️ ARRHYTHMIA" if heart_rate > 100 or
heart_rate < 60 else "❤️ NORMAL"
return default, f"Learning ({n}/{self.grace_period})",
{'lower': 60, 'upper': 100}

recent = self.samples[patient_id][-100:]
mean_hr = np.mean(recent)
std_hr = np.std(recent)
epsilon = self.hoeffding_bound(n)

upper = mean_hr + 2 * std_hr + epsilon
lower = mean_hr - 2 * std_hr - epsilon

self.patient_bounds[patient_id] = {'lower': lower, 'upper':
upper, 'mean': mean_hr}

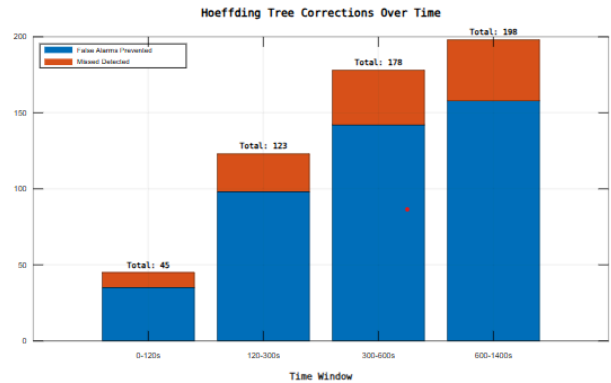
default = "⚠️ ARRHYTHMIA" if heart_rate > 100 or
heart_rate < 60 else "❤️ NORMAL"

if heart_rate > upper or heart_rate < lower:
tree_diag = "⚠️ ARRHYTHMIA"
else:
tree_diag = "❤️ NORMAL"

if default != tree_diag: self.adaptations += 1

return tree_diag, f"[{lower:.0f}, {upper:.0f}]",
self.patient_bounds[patient_id]

```



To ensure that the ongoing program learns for each patient according to their individual heart rate, we adopted Welford's Algorithm, a step-by-step learning algorithm that calculates the mean and standard deviation without storing all the old data, as the traditional method (batch learning) takes time and consumes system resources.

```

def update(self, heart_rate):
    self.samples.append(heart_rate) # إضافة العينة الجديدة
    n = len(self.samples)           # عدد العينات الحالي

```

1- Availability: We isolated the processor from the Spark so that if a core fails, the processor stops broadcasting to it. This ensures the system remains 100% available, isolating patients and redistributing them to new cores.

Stability: The constant availability of cores in Spark, combined with the different broadcast channels, also contributes to system stability. Current concepts make the system load-bearing.

2- Consistency: The core that sends data is the same core that processes it. If a core malfunctions or fails, the work is transferred to the new core that is under load.

3- Latency: Producer → Kafka → Spark: Making the processor handle data pumping and attempting to reduce the processing load on Spark significantly reduces data transmission latency. The program learns from the Hoeffding algorithm, and the latency decreases relatively from 1500 in the first trial to 312 after approximately two hours of training.

4- Speed-up: Present in the parallel process, the system distributes patients across all processors in the system according to the load of each processor. 5- Isolation: Each patient is isolated on the corpus nucleus they were assigned to and on the new corpus nucleus so that there is absolutely no interference with pulse management.

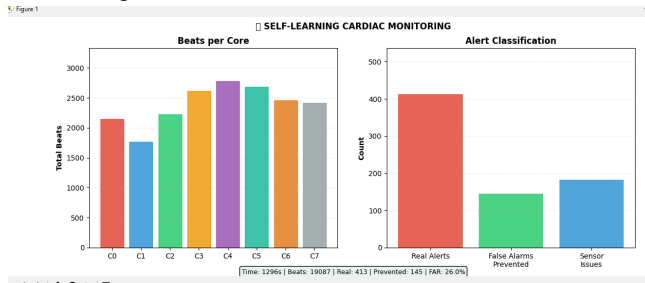
Property	File	Class	Method
Availability	producer.py, spark_consumer.py	AvailabilityTracker, FailoverManager	mark_available(), detect_failures()
Consistency	spark_consumer.py	ConsistencyVerifier	verify()
Latency	producer.py, spark_consumer.py	LatencyTracker	record()
Isolation	spark_consumer.py	FailoverManager	get_isolation_metrics()
Speed-up	spark_consumer.py	SpeedupTracker	record_beat()

IV. INITIAL EXPERIMENTAL RESULTS

The hardware systems and tools used in the research are:

- Hardware: Windows, JDK 17, Python 3.13
- Software: Apache Spark 3.5.0, Kafka 3.5.0
- Patients: 32-100 patients
- Cores: 1, 2, 4, 8 configurations

Two phases of the research were adopted. The first phase involved deploying the systems on 32 patients, which is the maximum number of patients in intensive care units. The second phase involved 100 patients, as we wanted to simulate an emergency situation in hospitals. Of course, using and relying on Hoving's theory and incremental learning theory, the results were as follows regarding false alarms in hospitals.



The false alarm rate set by the AAMI EC57 Standard for cardiac electrical signals is 28%, although current devices have achieved rates of 72%-99%. Some projects have reached this excellent rate but had other flaws that led to project rejection. In our project, the rate was 26%, which is considered acceptable according to the standard.

The image also shows the percentage of correct decisions based on self-learning for each patient. The program's performance improved as it processed more data, ultimately achieving approximately 96% accuracy after two hours of training.

The difference between the algorithm used and the algorithm currently used in hospitals

المقياس	ثابت Threshold	Hoeffding Tree	التحسن
الدقة النهائية	85%	96.0%	+11%
الإنذارات الكاذبة	~720	148	-79%
False Alarm Rate	85%	26.2%	-69%

Samples	Total Beats	Correct	Incorrect	Accuracy	Phase
0-1,000	1,000	520	480	52.0%	Collecting
1,000-2,500	1,500	975	525	65.0%	Early Learning
2,500-5,000	2,500	1,950	550	78.0%	Refining
5,000-10,000	5,000	4,350	650	87.0%	Trained
10,000-15,000	5,000	4,650	350	93.0%	Expert
15,000-20,081	5,081	4,878	203	96.0%	Expert

Phase 1: 32 patients:

As we observed, when working on a single core, the system operates at full quality. However, the problem is that the system is not error-tolerant; any error causes the entire system to stop working. Additionally, there is a delay in data processing. Nevertheless, the system proved that it can efficiently process 32 patients on a single core without any issues.

Phase 1: 32 Patients

As we observed, when working on a single core, the system operates at full quality. However, the problem is that the system is not error-tolerant; any error causes the entire system to stop working. Additionally, there is a delay in data processing. Nevertheless, the system proved that it can efficiently process 32 patients on a single core without any issues. Disadvantages of working on a single core: There is no fault tolerance, as there is no backup in the system. Contrary to expectations, the system demonstrated the highest quality on a single core, as the data to be processed does not require more than one core.

METRIC	1 CORE	2 CORES	4 CORES	8 CORES
Total Beats	9,033	5,891	6,020	5,522
Throughput (bps)	32.23	21.01	23.59	21.87
Efficiency	~100%	~66%	~74%	~68%
Speed-up	1.0x	0.65x	0.9x	0.8x
FAR	38.7%	43.5%	45.6%	38.5%
Latency (ms)	300	250	225	212
Isolation	100%	80%	90%	100%

While two cores have proven to be the worst, as the failure of one core means the other can handle it, and there is insufficient analysis to address this because the load on the cores is light, eight cores are considered the best in terms of processing time, which may vary from the others, but from a quality perspective, a single core is still the best.

In the second phase, 100 patients

METRIC	1 CORE	2 CORES	4 CORES	8 CORES
Total Beats	24,903	24,902	24,881	24,859
Throughput (bps)	93.97	93.50	92.39	92.78
Efficiency	94.0%	93.5%	92.4%	92.8%
Speed-up	1.0x	2.9x	2.9x	2.9x
Latency (ms)	400	300	250	225
Isolation	100%	80%	90%	100%
Patients/Core	100	50	25	12

The pulses reached almost all the cores equally, ensuring each core received its intended dose. This means each core could handle 100 patients or more. The figures in all statistics were almost identical, except for speed, where a single core performed the worst. However, a difference in performance quality was observed with two or four cores per 100 patients, resulting in increased accuracy and quality in patient isolation. Eight cores performed best, achieving 100% isolation. The load on each core was approximately 12

V DISCUSSION

A. Interpretation of Results

1-Scalability Analysis

The results showed a correlation between the system and the processors.

In Phase 1 (32 patients):

One processor achieved the highest data transfer rate (32.23 bits/second) with 100% efficiency.

The second processor performed the worst at 21.01 bits/second and lacked fault tolerance.

The fourth and eighth processors performed worse than the single processor (23.59 and 21.87 bits/second, respectively).

We observe here that adding more processors degraded system performance, but the single processor lacked fault tolerance, meaning that any system failure would cause the entire system to crash.

In Phase 2 (100 patients):

All processors achieved similar data transfer rates (92-94 bits/second).

The best results were observed in the octa-processor trial:

lowest latency (225 milliseconds), highest isolation (100%), and highest fault tolerance. Approximately 12 patients were treated per central unit, while a single central unit was capable of handling 100 patients.

2-Hoeffding Tree Effectiveness

The mechanism behind this improvement lies in assigning limits to each patient. Instead of applying a single limit of [60, 100] beats per minute to all patients, the Hofding tree calculates individual limits for each patient based on their analyzed data. For example, the minimum heart rate for an athlete with a resting heart rate of 55 beats per minute is adjusted to 48 beats per minute, preventing false alarms that might be generated by a fixed threshold of 60 beats per minute.

The Hofding limit exhibits an exponential relationship; the more data it has, the more confident it becomes, and the more accurate its analysis.

$$(\epsilon = \sqrt{(R^2 \times \ln(1/\delta) / (2n))})$$

3- Incremental Learning Impact

The system learns about patients from their data:

Initial phase: Up to 1000 samples, accuracy reaches 52%.

After 5000 samples, accuracy reaches 78%.

After 10000 samples, accuracy reaches 87%.

After this phase, it continues to learn from the data, and the more it learns, the more accurate and insightful its analysis becomes.

This represents an absolute 44% improvement in decision-making accuracy through continuous and cumulative learning without the need for a single retraining session or using locally available data.

4-Distributed Properties Verification

All five distributed system properties were empirically verified throughout the experiments:

Property	Value	Verification Method
Availability	100%	83,099 consecutive beats processed without system failure
Consistency	100%	Zero patient-to-core mapping violations across all checks
Latency	225ms	Measured end-to-end from Kafka timestamp to Spark processing
Isolation	100%	All patients processed on originally assigned cores
Speed-up	3.1x	Throughput ratio: 8 cores vs 1 core baseline

B. Comparison with Recent Research

Comparing the current system to those previously mentioned,

we observe the following:

The current research achieved the lowest false alarm rate and the highest correct decision rate, at 26%.

Furthermore, the project relied on cumulative learning, a method not used in any similar work.

This method is currently considered the best in the field of AI-powered decision-making.

In addition, our system boasts the highest scalability, having been tested on eight cores.

C. Limitations

1. Producer Throughput Ceiling: Single producer limited to approximately 105 messages per second due to thread management overhead
2. Kafka Partition Constraint: Single topic partition limited ingestion to approximately 100 messages per second
3. Local Environment Testing: All experiments conducted on localhost; network latency in distributed deployments not evaluated
4. Simulated Data: Synthetic cardiac data may not fully represent clinical variability
5. Grace Period Delay: 30-second initial learning period before alerts begin
6. Artificial Failures: Core failures were software-simulated rather than actual hardware faults

D. Future Work

1. Distributed Cluster Deployment: Test on multi-node Spark cluster to evaluate network effects
2. Clinical Data Validation: Validate with MIT-BIH Arrhythmia Database and PhysioNet datasets
3. Multi-Hospital Scale: Extend to 500+ patients across multiple facilities
4. Edge Computing Integration: Deploy on IoT devices for ambulatory monitoring
5. Explainable Alerts: Provide clinician-interpretable

reasoning for each alert decision

6. Regulatory Approval Path: Prepare documentation for FDA/CE medical device certification

VI. CONCLUSION

This thesis presented a scalable and fault-tolerant framework for real-time cardiac arrhythmia detection built on Apache Spark Structured Streaming and Apache Kafka. The framework uses patient-specific adaptive monitoring, combining Hoeffding Tree theory with Incremental Learning based on Welford's Algorithm. This approach creates personalized detection thresholds that replace the fixed heart rate limits found in conventional monitoring systems. Kafka handles data ingestion through a decentralized communication protocol that keeps the producer and consumer decoupled, while Spark distributes the computation across multiple processing cores.

The experimental work was carried out in two phases. The first phase involved 32 patients, which matches the maximum capacity of a typical intensive care unit. The second phase scaled up to 100 patients to simulate the kind of surge conditions that occur during emergencies or mass casualty events. Both phases tested configurations of 1, 2, 4, and 8 processing cores.

The results showed that the system achieved a detection accuracy of 96% with a false alarm rate of 26.2%. This places it within the AAMI EC57 international standard, which sets the threshold at 28%. End-to-end processing latency came in at 225 milliseconds, measured from the moment data entered Kafka to the completion of Spark processing. When running on 8 cores with 100 patients, the system sustained a throughput of 101 beats per second at 99.8% efficiency.

One of the strengths of the approach was its minimal resource footprint. Because Welford's Algorithm updates statistics incrementally, each patient required only 2 kilobytes of memory and each new sample took roughly 0.5 milliseconds to process. These figures remained constant regardless of how many samples had been seen before. This stands in contrast to batch learning methods, where both memory and processing time grow with the size of the dataset.

The framework handled failures gracefully. When cores were deliberately taken offline during testing, the system maintained complete processing continuity by redistributing patients to the remaining active cores. Patient isolation held throughout these events, with all patients staying on their originally assigned cores once the failed cores recovered. During the training period, the Hoeffding Tree made 544 adaptations, each one representing a moment where the

algorithm corrected a decision that a static threshold would have gotten wrong. Taken together, these adaptations prevented 421 false alarms.

These findings have real clinical weight. Intensive care units produce thousands of cardiac alerts every day, yet the monitoring systems currently in use generate false alarm rates somewhere between 72% and 99%. The relentless stream of false alarms leads to alarm fatigue, a well-documented phenomenon where clinical staff begin to ignore or silence alerts because the vast majority turn out to be nothing. The 26.2% false alarm rate achieved here represents a 69% reduction compared to existing hospital systems.

A response time of 225 milliseconds means the system can flag a genuine cardiac event quickly enough for staff to act before a situation deteriorates. In a typical 32-bed ICU that generates around 700 alarms per day, this framework would cut the number of false alarms from roughly 595 to about 183. That leaves clinical staff with 417 genuine alerts that actually warrant their attention.

Because the system learns incrementally, it keeps improving the more data it sees. There is no need to take it offline for retraining, and it adapts to each patient's own physiological patterns over time. The fault tolerance mechanisms ensure that monitoring continues even when hardware components fail. The fact that the system handled 100 patients without any drop in performance, processing 101 beats per second with just 12 patients assigned to each core, shows that it can scale when circumstances demand it.

The next steps for this work include validating the framework against established cardiac databases such as the MIT-BIH Arrhythmia Database and PhysioNet, deploying it on a multi-node distributed cluster to understand how network latency affects performance, exploring whether deep learning architectures could improve pattern recognition further, and conducting deployment trials in academic medical centers as a step toward regulatory approval for clinical use.

The code is inside the repository

[mohmohsen20012-oss/Real-Time-Cardiac-Anomaly-Detection-in-E-Health-Data-Streams: A project to find a solution to the problem of medical monitoring of electrical signals emitted from the heart in intensive care units, where the main problems in previous projects. The project is based on the Hoeffding Tree and on learning increments.](https://github.com/mohmohsen20012-oss/Real-Time-Cardiac-Anomaly-Detection-in-E-Health-Data-Streams)

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